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Computer Science

Numerical Linear Algebra

Handin 10

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May 3, 2026

I'm gonna be flipping burgers in 1 year ;_;

Anyway, consider the matrix

$$A = \begin{bmatrix} 0.5 & 0.25 & 0.25 \\ 0.25 & 0.3 & 0.45 \\ 0.25 & 0.45 & 0.3 \end{bmatrix} \tag{1}$$

(a) Verify that A is symmetric and explain why A is diagonalizable with only real eigenvalues

Verifying that A is symmetric can be done by looking at it and saying 'Yep that's symmetric', since $row_i == column_i$

Or verify it using python.

```
A = np.array([[0.5, 0.25, 0.25],
              [0.25, 0.3, 0.45],
              [0.25, 0.45, 0.3]])

if np.array_equal(A, A.T):
    print(True)
# Output for reference
True
```

Per the spectral theorem 23.10 [1] any real symmetric matrix $A \in \mathbb{R}^{n \times x}$ there exist an orthogonal matrix $V \in \mathbb{R}^{n \times n}$ such that $A = V \Lambda V^T$ where Λ is a diagonal matrix, and because we can factor the matrix (1) into that, then it can be said to be diagonalizable.
To go even further, the Theorem also states that A has n real eigenvalues and an orthonormal basis consisting of it's egenvectores...Yeah take that 7-year old Gauss! I'm smart too!

(b) Verify that 1.0 is an eigenvalue of A, and then determine the other eigen-values of A.

To verify it's possible to use proposition 24.4 [1] which says if A is a stochastic matrix, then 1.0 is an eigenvalue of A. A stochastic matrix is a square matrix $n \times n$ with non-negative entries and the sum of each row is 1.0. Looking at the matrix (1) it's trivial that the sum of each row is 1.0, so 1.0 is an eigenvalue of A.

To find the rest of the eigenvalues needed to construct the basis for A, the characteristic

polynomial for A is used:

$$p(\lambda) = \det(A - \lambda I_3) = \det \begin{bmatrix} 0.5-\lambda & 0.25 & 0.25 \\ 0.25 & 0.3-\lambda & 0.45 \\ 0.25 & 0.45 & 0.3-\lambda \end{bmatrix}$$
$$= (0.5-\lambda)\det \begin{bmatrix} 0.3-\lambda & 0.45 \\ 0.45 & 0.3-\lambda \end{bmatrix} - (0.25)\det \begin{bmatrix} 0 & 0.15-\lambda \\ 0.25 & 0.3-\lambda \end{bmatrix} + (0.25)\det \begin{bmatrix} 0.25 & 0.3-\lambda \\ 0.25 & 0.45 \end{bmatrix}$$

Calculating the determinants and simplifying gives us the characteristic polynomial:

$$p(\lambda) = -\lambda^3 + 1.1\lambda^2 - 0.35\lambda + 0.05 \tag{2}$$

(c) Compute an orthonormal basis for $\mathbb{R}^3$ consisting of eigenvectors of A.

(d) Determine an orthogonal matrix V and a diagonal matrix Λ such that

$$A = V\Lambda V^T \tag{3}$$

(e) For

$$v = \begin{bmatrix} 0.12 \\ 0.21 \\ 0.39 \end{bmatrix} \tag{4}$$

Use the results to determine the limits of $A^k v$ for $k \to \infty$

Appendix

References

[1] Andrew Swan. *Numerisk Lineær Algebra*. Institut for Matematik, Aarhus Universitet, 2025.